Amadeo De La Vega

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Education

2022-present **Doctor of Philosophy in Computer Science**, *University of Maryland (UMD)*, College Park

2019-2021 **Master of Science in Mathematics**, *National Autonomous University of Mexico (UNAM)*, Mexico City
GPA: 96/100

2011–2016 **Bachelor (Hons) of Science in Mathematics**, *Autonomous University of State of Morelos (UAEM)*, Cuernavaca, Morelos
GPA: 97/100

Relevant Professional Experience

Spring 2025-present **Full Teaching Assistant**: CMSC 330 (Organization of Programming Languages)

Fall 2024 **Full Teaching Assistant**: CMSC 250 (Discrete Structures)

Spring 2024 **Full Teaching Assistant**: Organization of Programming Languages

Fall 2023 **Full Teaching Assistant**: Discrete Structures

Spring 2023 **Full Teaching Assistant**: CMSC 858L (Quantum Complexity Theory)

Fall 2022 **Full Teaching Assistant**: CMSC 456 (Cryptography)

Jun. 2021-Present **Member of QMexico**: Organize Quantum Computing events and Create educational resources for researchers, professionals, students, and enthusiasts worldwide. `https://qmexico.org/our-team/`

Jul. 2021-Aug. 2021 **Spanish tutorial session leader** of the Quantum Summer School 2021: Taught international students Quantum Computing and how to use Qiskit and Cirq (IBM and Google quantum programming software resp.) `https://qworld.net/quantum-summer-school-2021/`

Jun. 2018-Aug. 2018 **Electoral Assistant Trainer** (CAE) of the National Electoral Institute (INE): Taught citizens how the Mexican elections work and Organized electoral activities.

Nov. 2017-May. 2018 **Programmer and mathematician** in Automotive Center Futurama SA. De CV: Worked on mathematical modeling, the development of algorithms, and diverse projects.

Sep. 2016-Jun. 2017 **Mathematics teacher**, Montessori de la Montaña: Taught algebra, geometry, and calculus at the high school level.

Oct. 2013-Sep. 2016 **Undergraduate research assistant** of the CONACYT. Prepared technical material for seminars required by Dr. Yuri Karlovich.

Research Experience

Fall 2025 Project: **"A new verifier for coding agents"**

Fall 2024	Project: **"Enhanced Security Analyzer System for AI-agents with Logical Operators"**
Fall 2024	Project: **"Multi-party Computation based Location Services"**
Spring 2023	Project: **"Deep Learning Approach to Malicious Network Traffic Classification"**
Fall 2022	Project: **"Inapproximability of Maximum Hamiltonian Path-S"**
2019–2021	Master's thesis in Quantum Complexity Theory: **"The PCP theorem into the quantum world"**
2013–2016	Undergraduate joint research project and thesis: **"Factorization of almost periodic matrix functions and its applications"**
Oct. 2014	Oral speaker at the XLVII National Congress of the Mexican Mathematical Society with a talk entitled **"Group theory and linear representations: Development of an algorithm to solve the Rubik's cube"**, *UJED, Durango*

Scholarships and Honors

Sep. 2021	Fulbright- García Robles scholarship for Graduate Studies alternate holder, awarded by the U.S. – Mexico Commission for Educational and Cultural Exchange (COMEXUS), Mexico City
2019–2021	National master's scholarship awarded by the National Council of Science and Technology (CONACYT), Mexico City
Aug. 2018-Dec. 2018	COMEXUS scholarship for the Long term English program: "International Graduate Preparation Program, Global Launch". ASU, United States
2013–2016	Research Assistant Scholarship awarded by the National Council of Science and Technology (CONACYT), Cuernavaca, Morelos
Dec. 2016	Bronze medal at the XIX Iberoamerican Mathematical Olympiad for University Students (OIMU). CIMAT, Guanajuato
Dec. 2015	Honorable mention at the XVIII Iberoamerican Mathematical Olympiad for University Students (OIMU). CIMAT, Guanajuato
Oct. 2015	Recognition for being assistant at the XLVIII National Congress of the Mexican Mathematical Society. UNISON, Sonora
Sep. 2015	Bronze medal at the VII Iberoamerican Mathematics Competition for University Students (CIIM). UNAM, Mexico City
2013–2015	Salary scholarship. UAEM, Cuernavaca Morelos
Dec. 2015	Recognition for Best GPA of generation 2011. UAEM, Cuernavaca, Morelos
Oct. 2014	Recognition for being an Oral speaker at the XLVII National Congress of the Mexican Mathematical Society, UJED, Durango

Technical Knowledge

- Python
- Lean
- Linux
- HTML/CSS/JavaScript
- Cirq
- LaTex
- Qiskit
- W. Mathematica
- OCaml
- Maple
- C